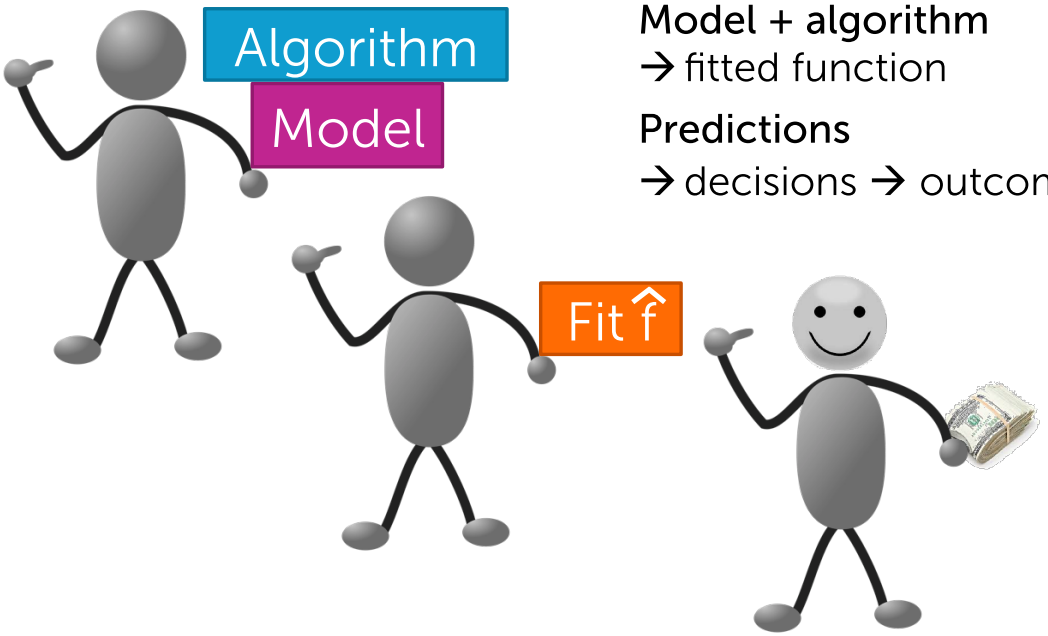
Make predictions, get $, right??



Model + algorithm
→ fitted function

Predictions
→ decisions → outcome

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Or, how much am I losing?

Example: Lost $ due to inaccurate listing price

- Too low → low offers
- Too high → few lookers + no/low offers

How much am I losing compared to perfection?

Perfect predictions: Loss = 0

My predictions: Loss = ???

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Measuring loss

Loss function:

$L(y, f_{\hat{w}}(\mathbf{x}))$

Cost of using $\hat{w}$ at x when y is true

actual value

$\hat{f}(\mathbf{x})$ = predicted value $\hat{y}$

Examples:

(assuming loss for underpredicting = overpredicting)

Absolute error: $L(y, f_{\hat{w}}(\mathbf{x})) = |y - f_{\hat{w}}(\mathbf{x})|$

Squared error: $L(y, f_{\hat{w}}(\mathbf{x})) = (y - f_{\hat{w}}(\mathbf{x}))^2$

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"Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful." George Box, 1987.

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Assessing the loss

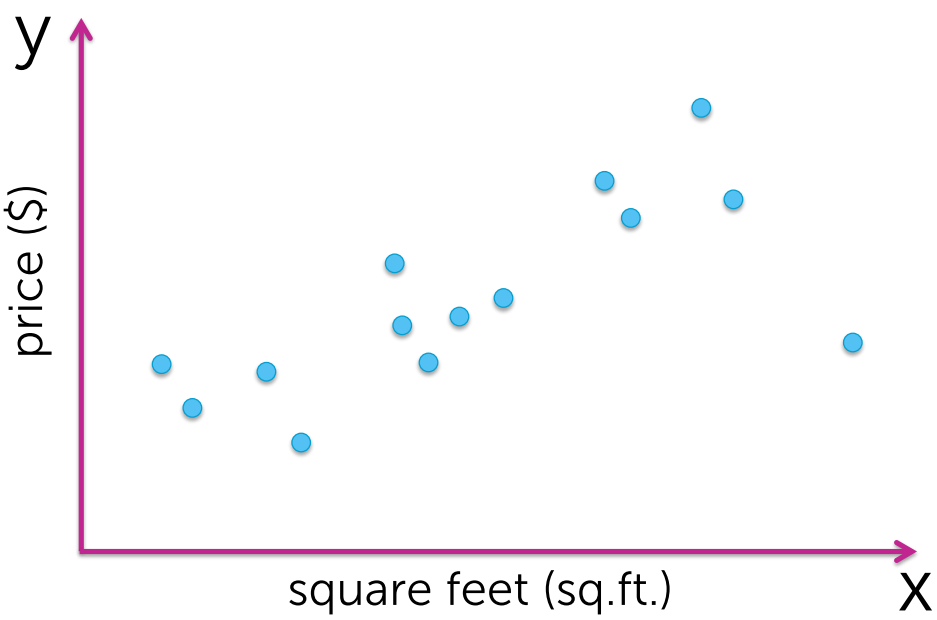
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Assessing the loss
Part 1: Training error

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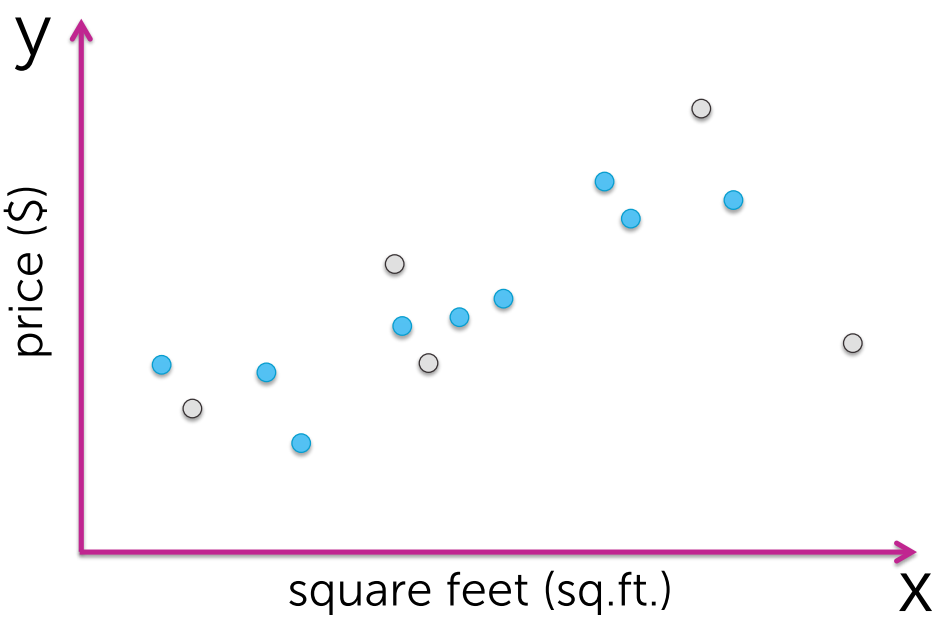
Define training data



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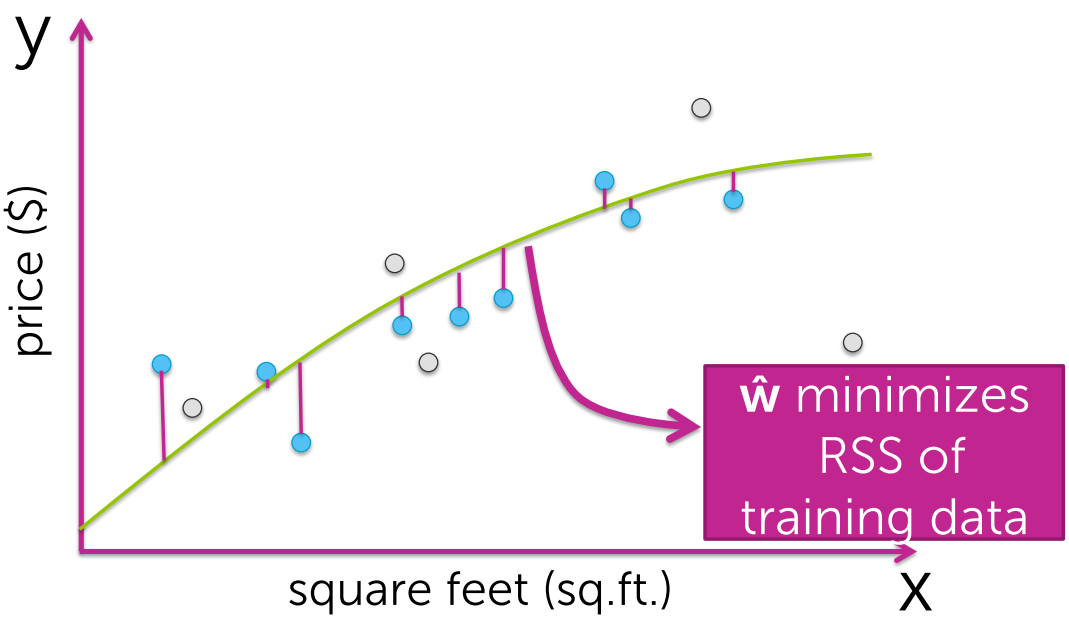
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Define training data



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Example:
Fit quadratic to minimize RSS



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Compute training error

1. Define a loss function $L(y, f_{\hat{\mathbf{w}}}(\mathbf{x}))$
   - E.g., squared error, absolute error,…

2. Training error

   = avg. loss on houses in training set

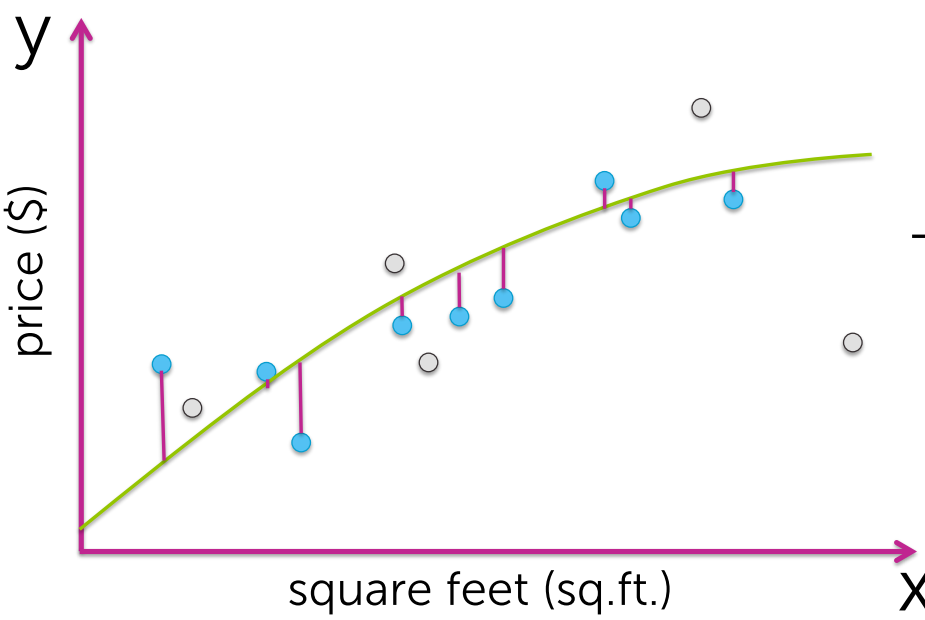
   $$= \frac{1}{N} \sum_{i=1}^{N} L(y_i, f_{\hat{\mathbf{w}}}(\mathbf{x}_i))$$



fit using training data

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Example:
Use squared error loss $(y-f_{\hat{w}}(x))^2$

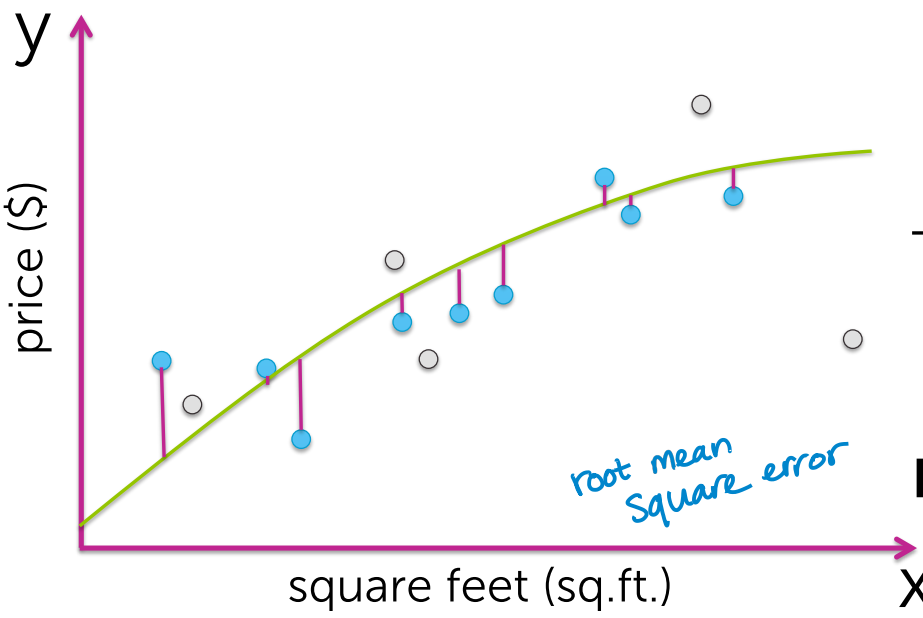


Training error ($\hat{w}$) = 1/N *
[($\$_{train\ 1}$-$f_{\hat{w}}$(sq.ft.$_{train\ 1}$))^2
+ ($\$_{train\ 2}$-$f_{\hat{w}}$(sq.ft.$_{train\ 2}$))^2
+ ($\$_{train\ 3}$-$f_{\hat{w}}$(sq.ft.$_{train\ 3}$))^2
+ … include all
training houses]

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Example:
Use squared error loss $(y-f_{\hat{w}}(x))^2$

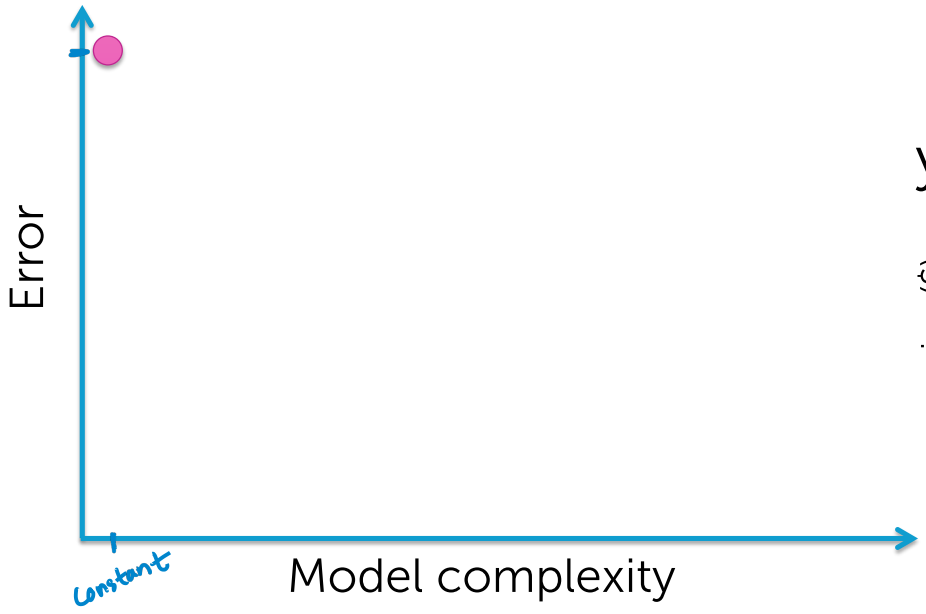


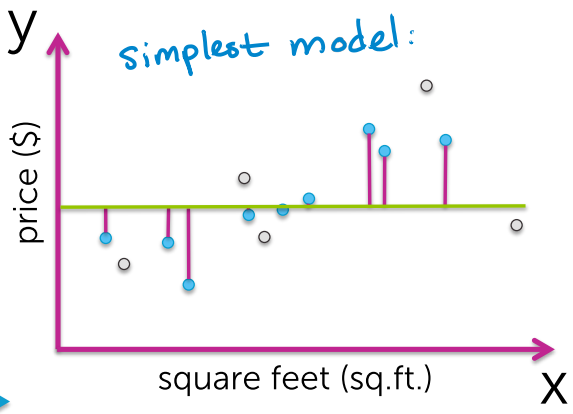
Training error ($\hat{w}$) = $\frac{1}{N}\sum_{i=1}^{N}(y_i-f_{\hat{w}}(x_i))^2$

RMSE = $\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-f_{\hat{w}}(x_i))^2}$

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Training error vs. model complexity

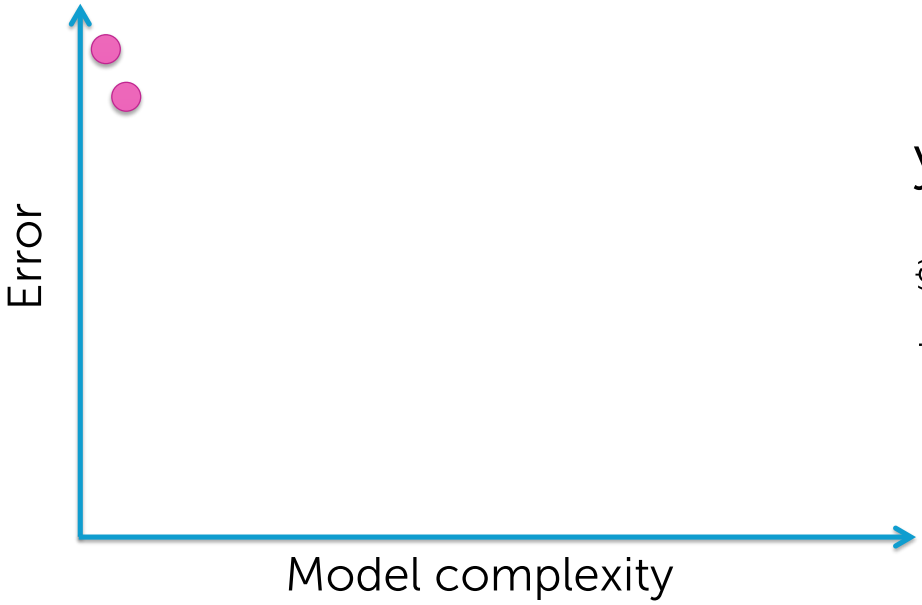


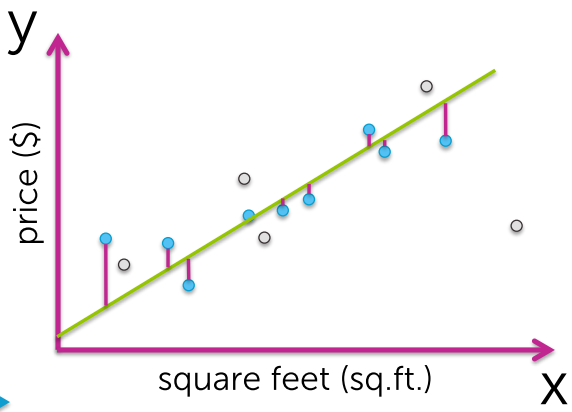


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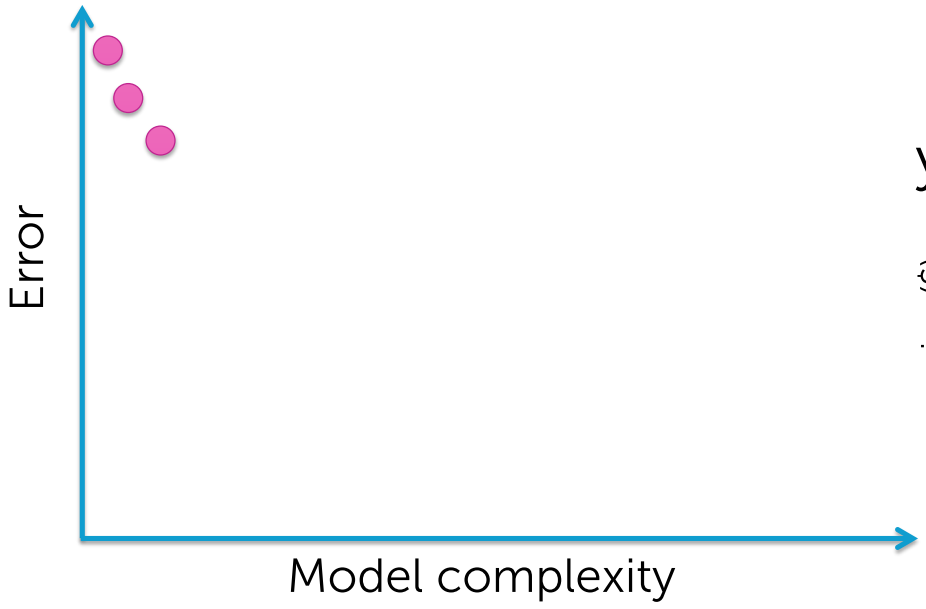
Training error vs. model complexity

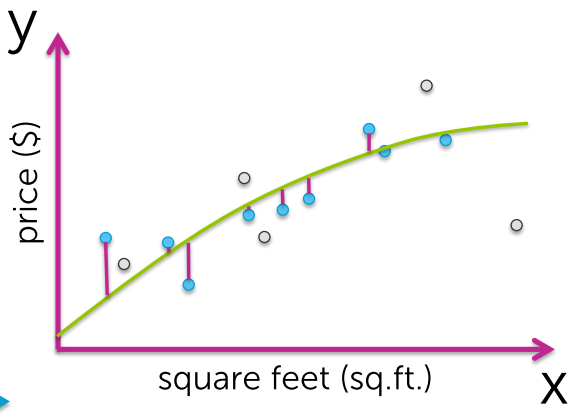




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Training error vs. model complexity

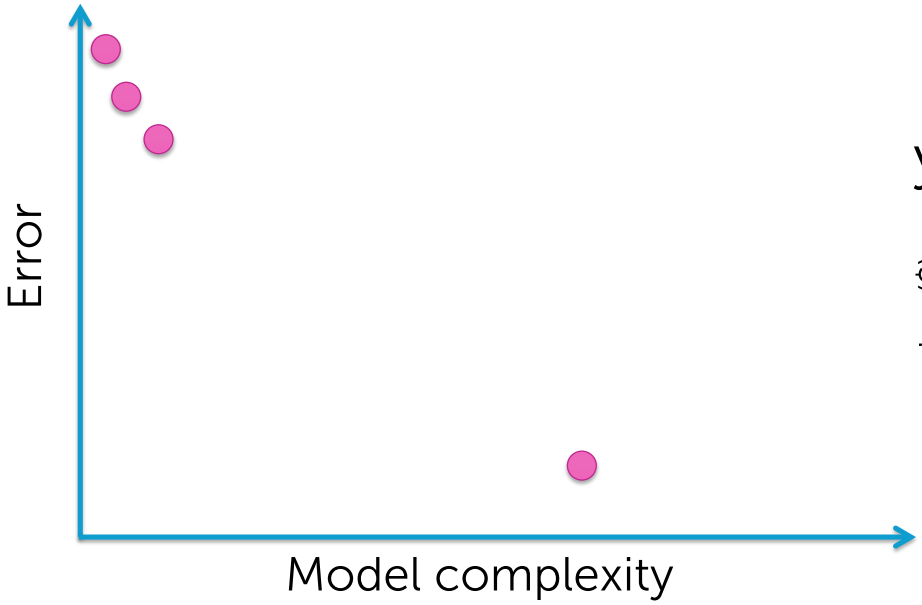


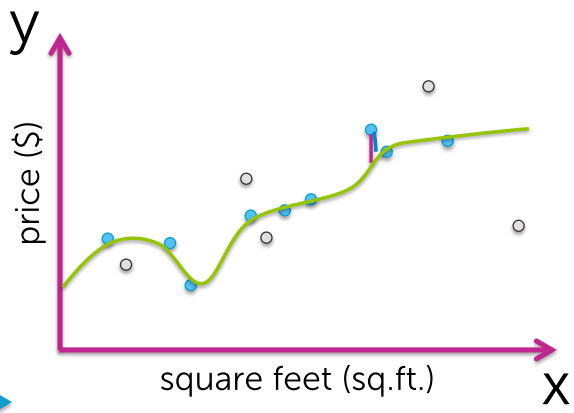


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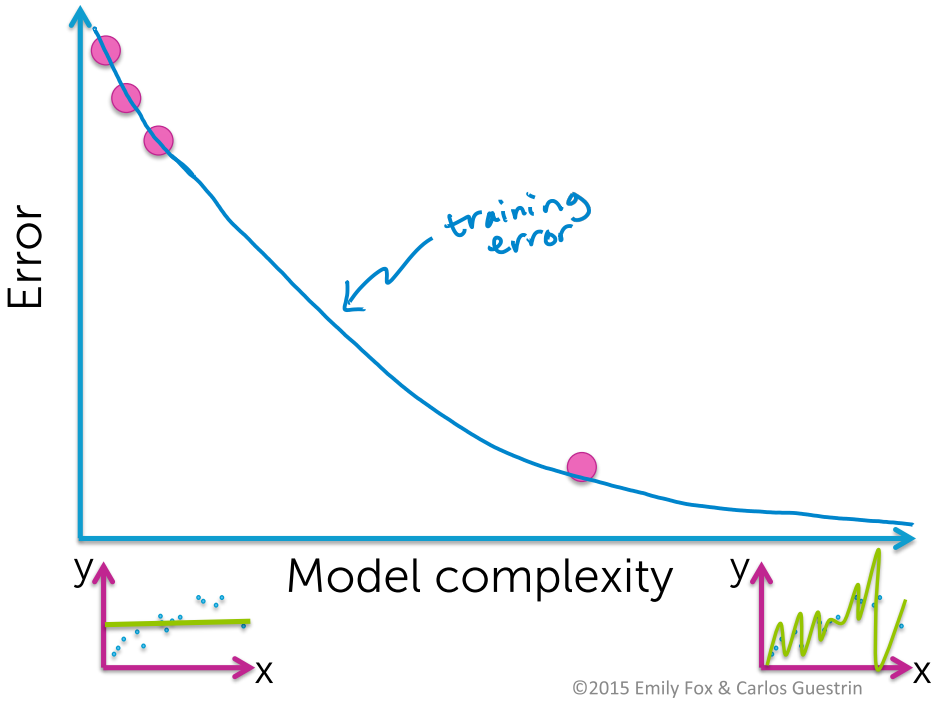
Training error vs. model complexity





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Training error vs. model complexity

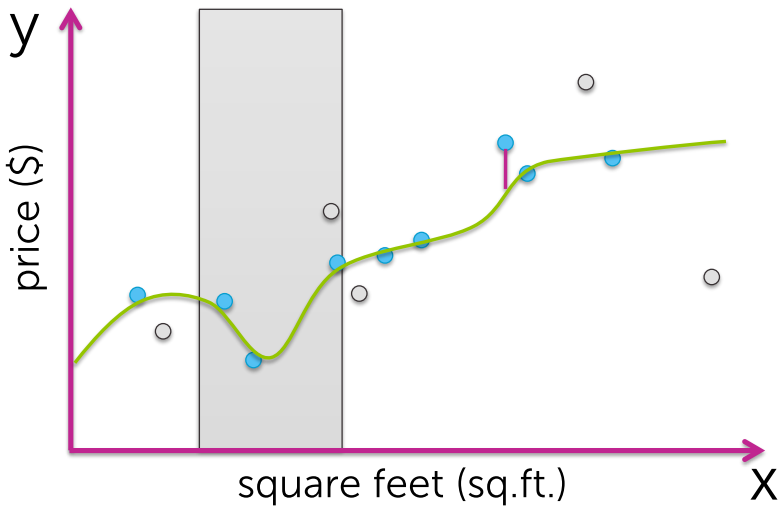


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Is training error a good measure of predictive performance?

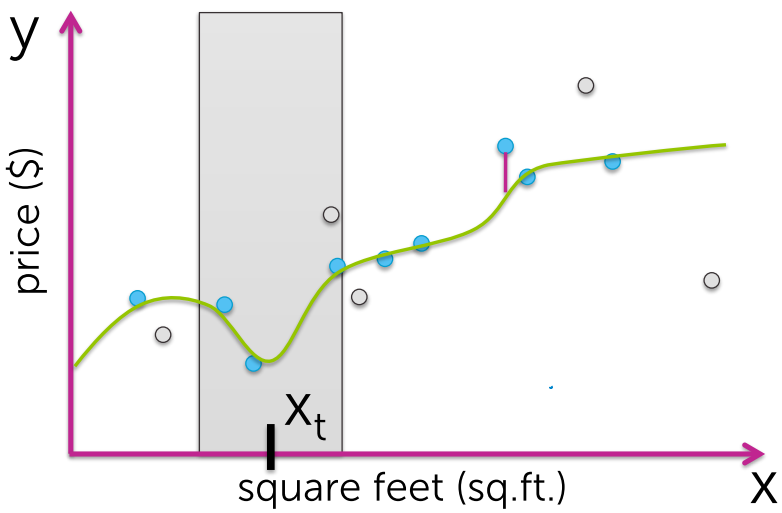
How do we expect to perform on a new house?



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Is training error a good measure of predictive performance?

Is there something particularly bad about having x_t square feet???



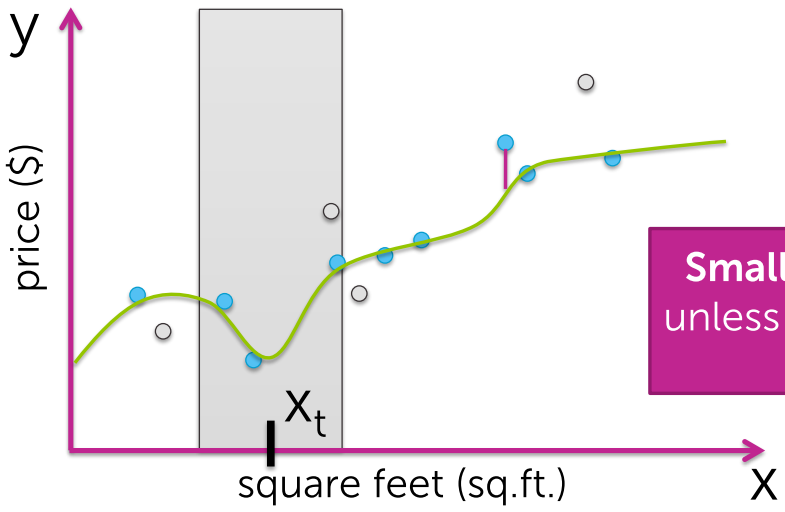
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Is training error a good measure of predictive performance?

Issue: Training error is overly optimistic

because $\hat{w}$ was fit to training data



Small training error ≠> good predictions
unless training data includes everything you might ever see

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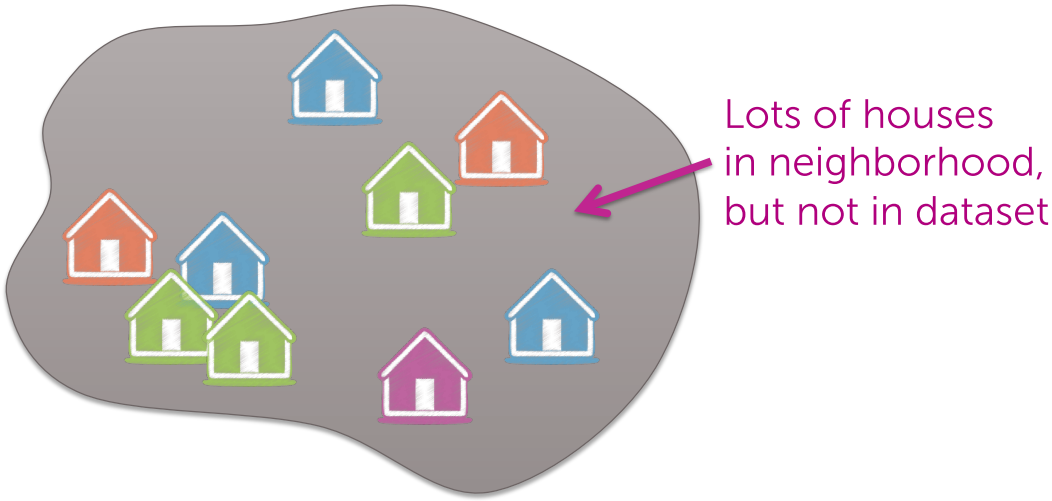
Assessing the loss
Part 2: Generalization (true) error

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Generalization error

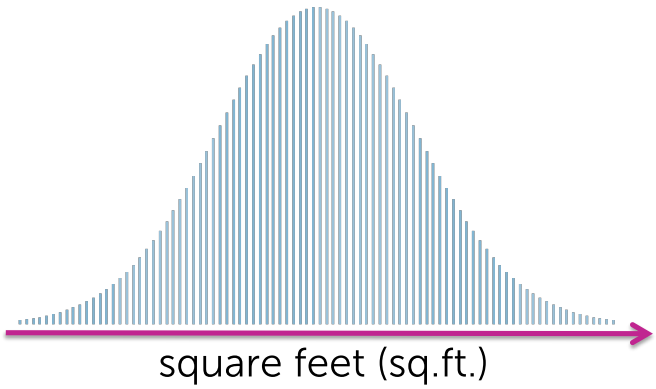
Really want estimate of loss over all possible (🏠,$) pairs



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Distribution over houses

In our neighborhood, houses of what # sq.ft. ( ) are we likely to see?



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Distribution over sales prices

For houses with a given # sq.ft. (),
what house prices $ are we likely to see?



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Generalization error definition

Really want estimate of loss over all possible (🏠,$) pairs

Formally:

average over all possible (**x**,y) pairs weighted by how likely each is

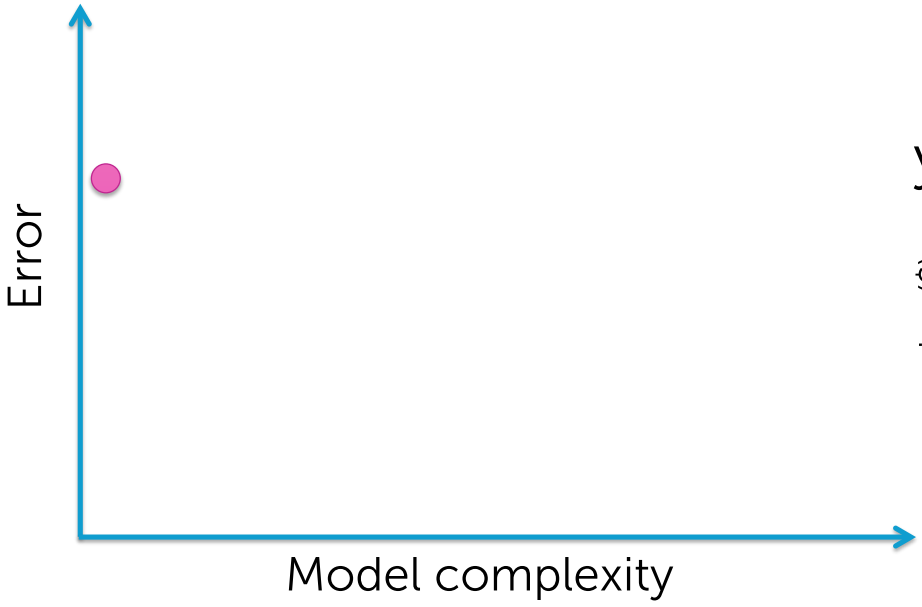
$$\text{generalization error} = E_{\mathbf{x},y}[L(y, f_{\hat{\mathbf{w}}}(\mathbf{x}))]$$

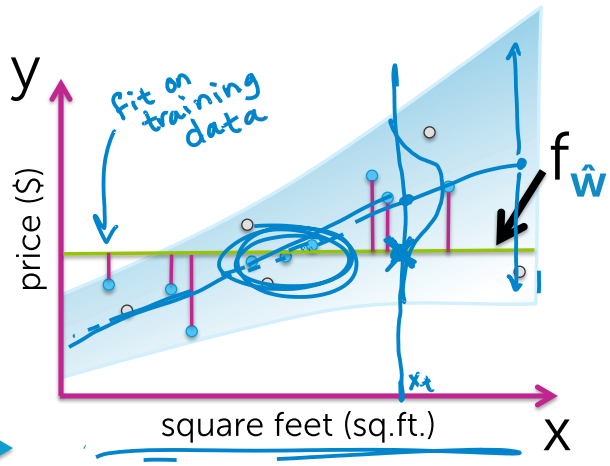
fit using training data

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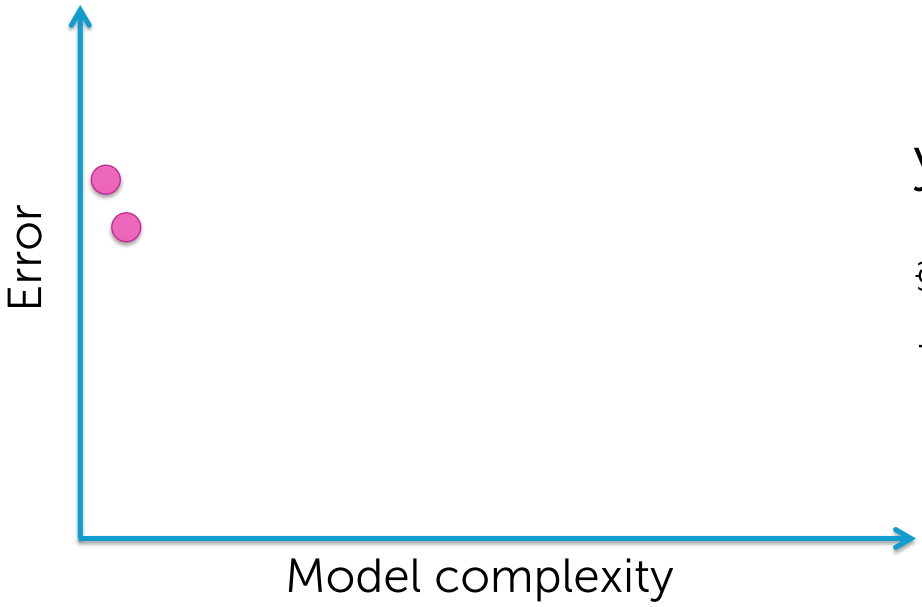
Generalization error vs. model complexity

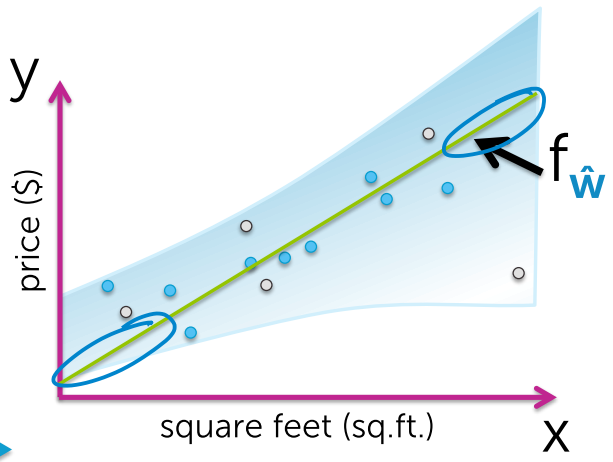




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Generalization error vs. model complexity

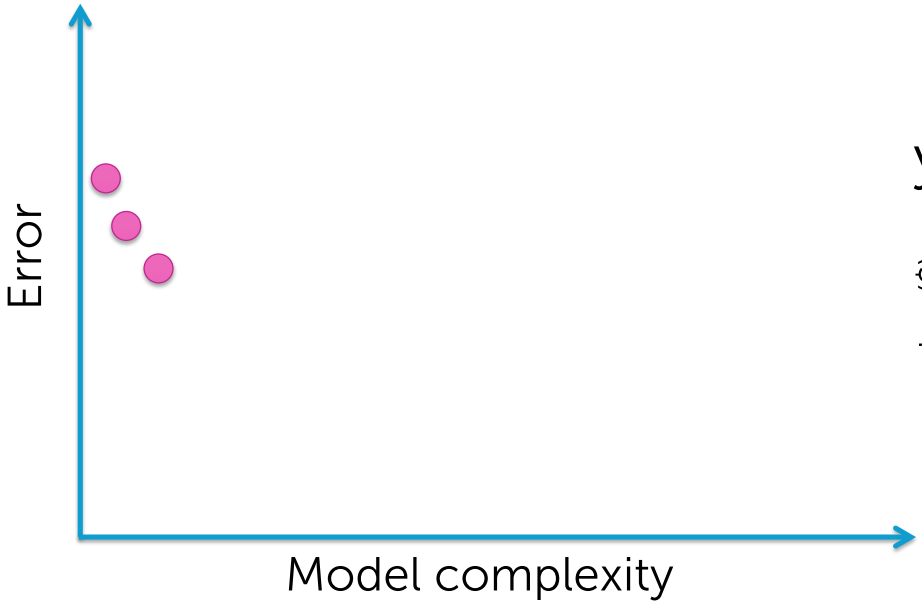


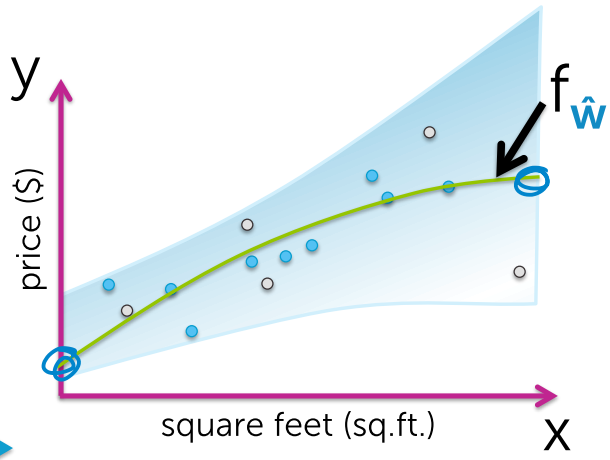


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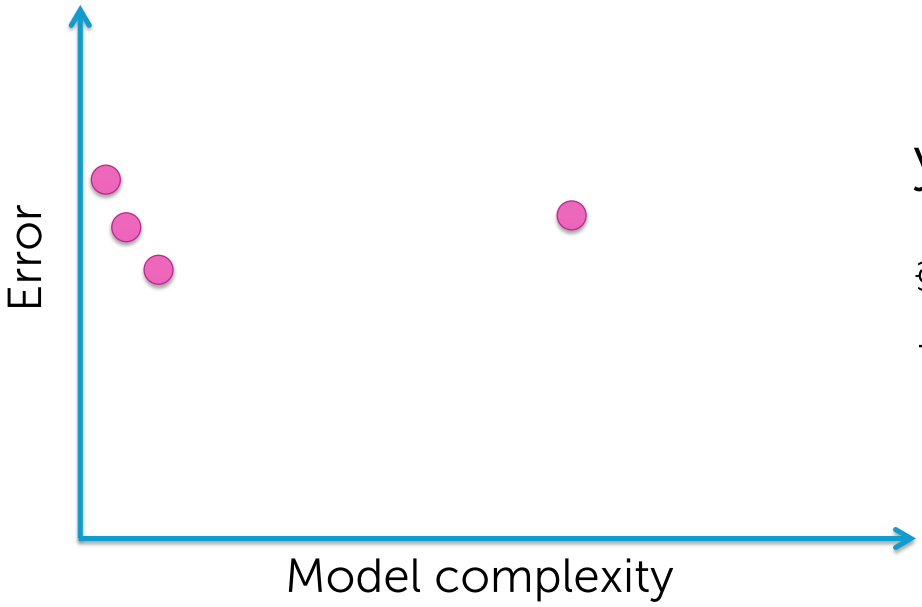
Generalization error vs. model complexity

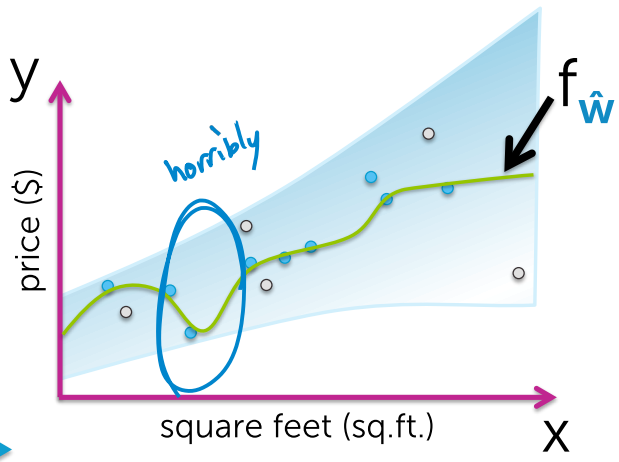




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Generalization error vs. model complexity

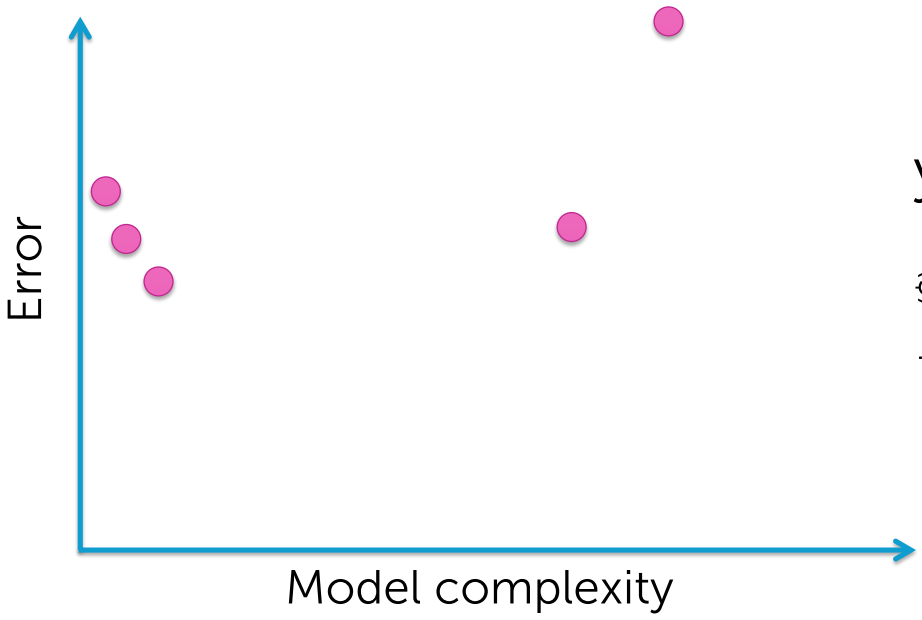


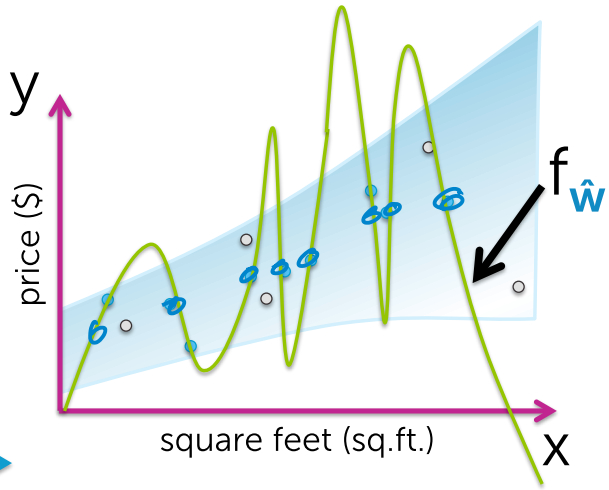


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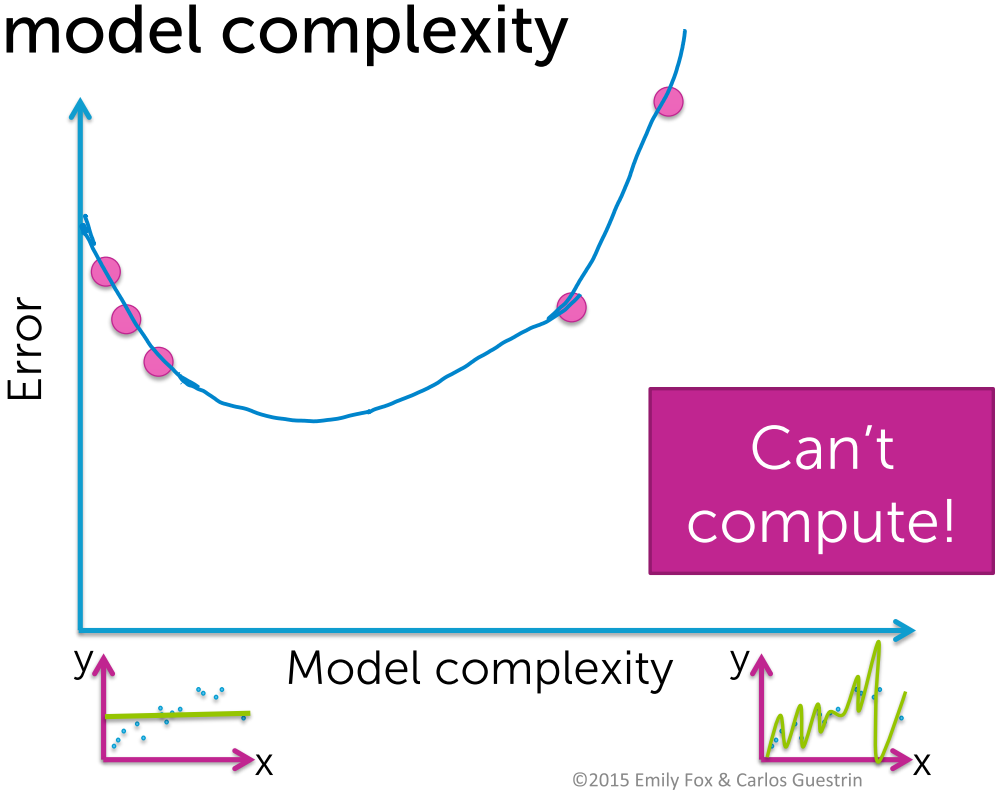
Generalization error vs. model complexity





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Generalization error vs. model complexity



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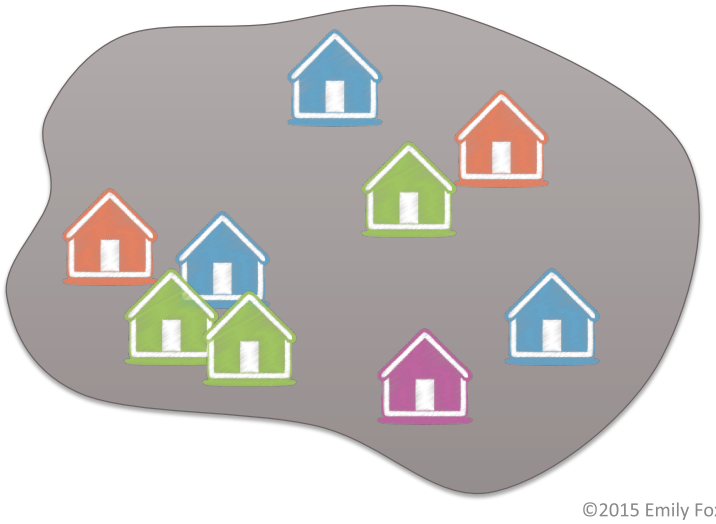
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Assessing the loss
Part 3: Test error

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Approximating generalization error

Wanted estimate of loss over all possible (🏠,$) pairs



Approximate by looking at houses not in training set

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Forming a test set

Hold out some (🏠,$) that are *not* used for fitting the model



Training set



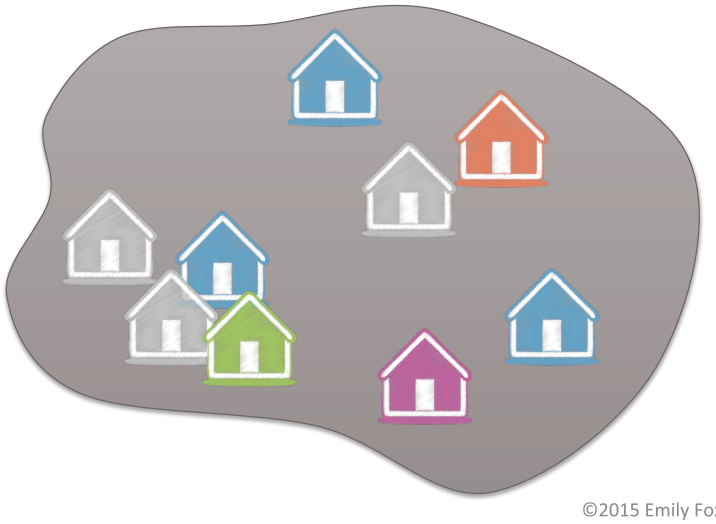
Test set



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Forming a test set

Hold out some ( ,$) that are *not* used for fitting the model



Proxy for "everything you might see"

Test set



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Compute test error

Test error

= avg. loss on houses in test set

$$= \frac{1}{N_{test}} \sum_{i \text{ in test set}} L(y_i, f_{\hat{w}}(\mathbf{x}_i))$$

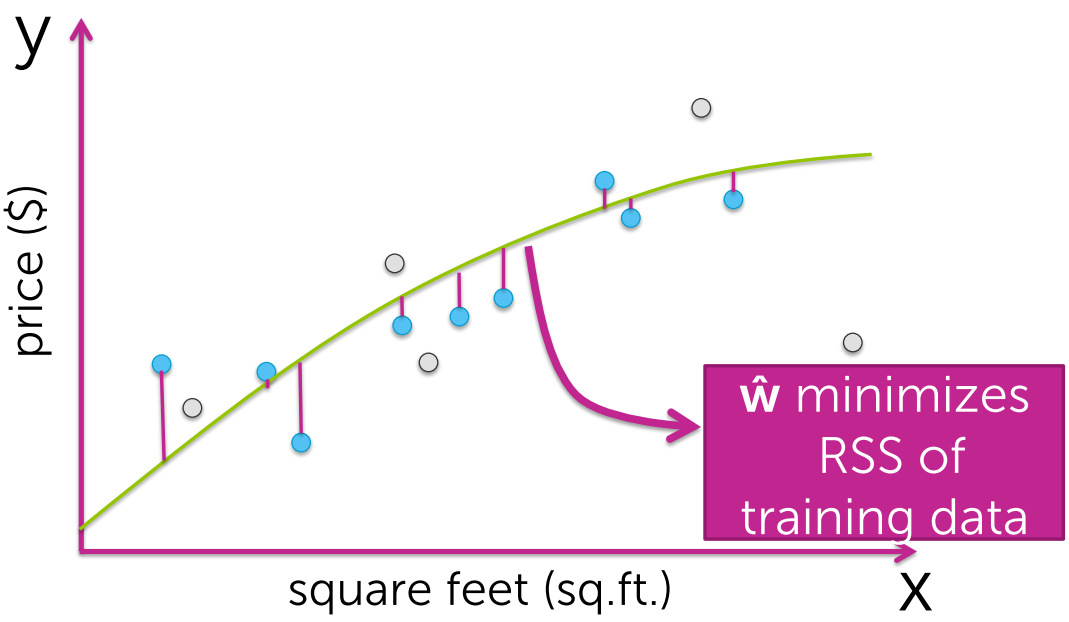
test points

fit using training data

has never seen test data!

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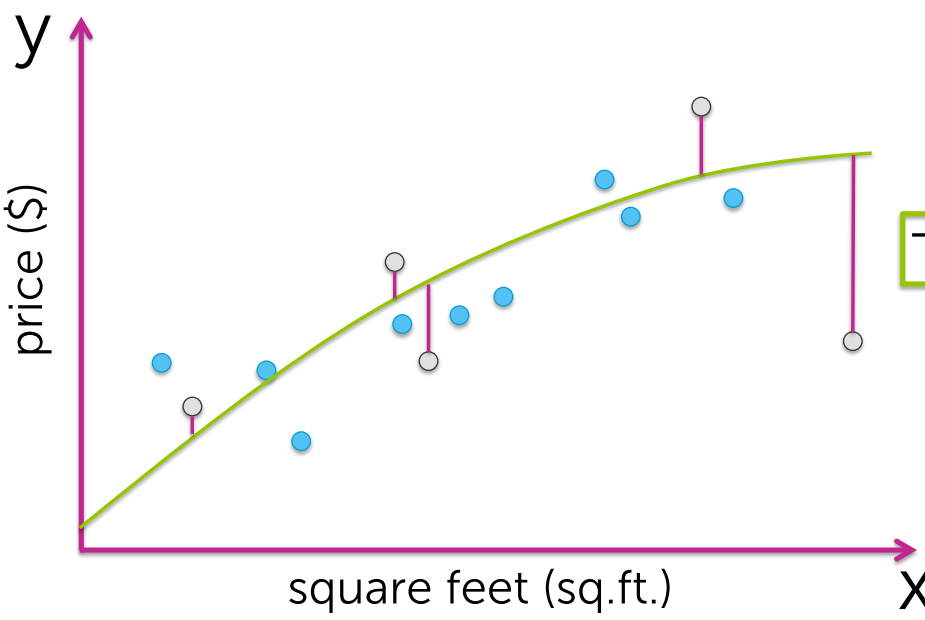
Example: As before, fit quadratic to training data



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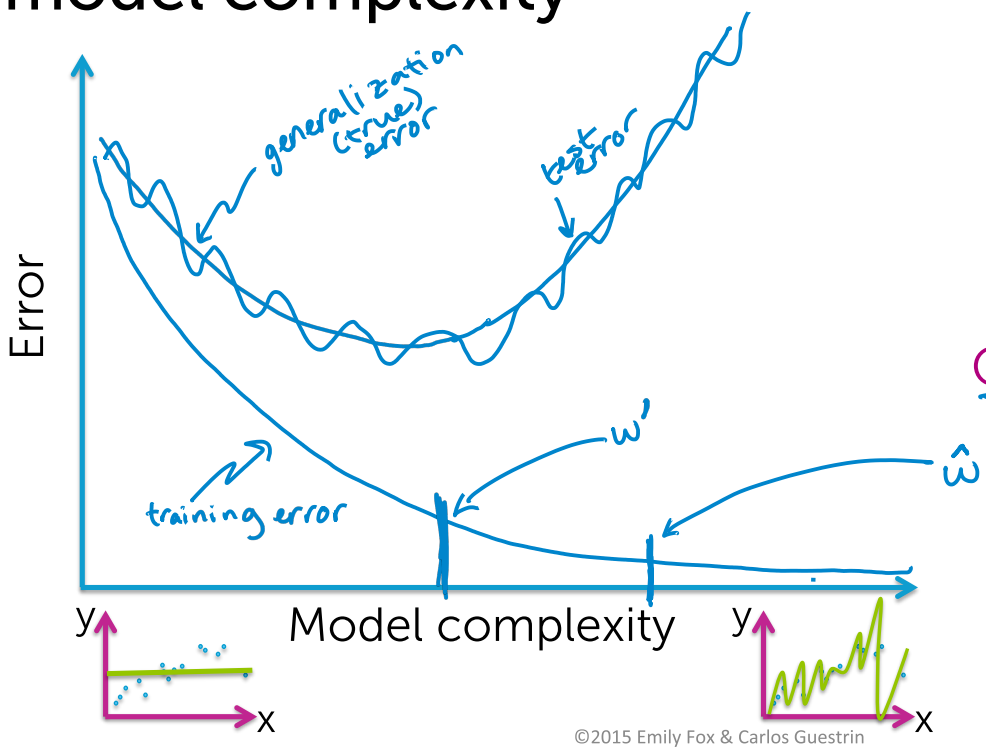
Example: As before, use squared error loss $(y-f_{\hat{w}}(x))^2$



Test error ($\hat{w}$) = 1/N *
[($\$_{test\ 1}$ - $f_{\hat{w}}(sq.ft._{test\ 1})$)^2
+ ($\$_{test\ 2}$ - $f_{\hat{w}}(sq.ft._{test\ 2})$)^2
+ ($\$_{test\ 3}$ - $f_{\hat{w}}(sq.ft._{test\ 3})$)^2
+ … include all test houses]

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Training, true, & test error vs. model complexity



Overfitting if:

If there exists a model with estimated params w' such that

1. training error (ŵ) < training error (w')
2. true error (ŵ) > true error (w')

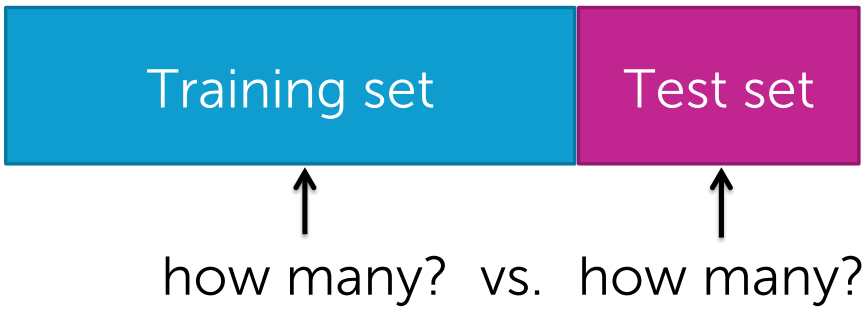
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Training/test split

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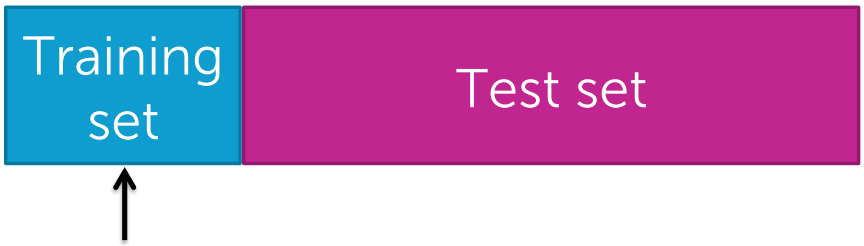
Training/test splits



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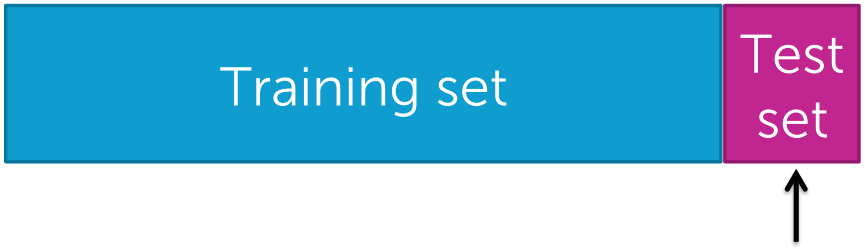
Training/test splits



Too few → $\hat{w}$ poorly estimated

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Training/test splits



Too few → test error bad approximation of generalization error

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Training/test splits



Typically, just enough test points to form a reasonable estimate of generalization error

If this leaves too few for training, other methods like **cross validation** (will see later…)

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3 sources of error + the bias-variance tradeoff

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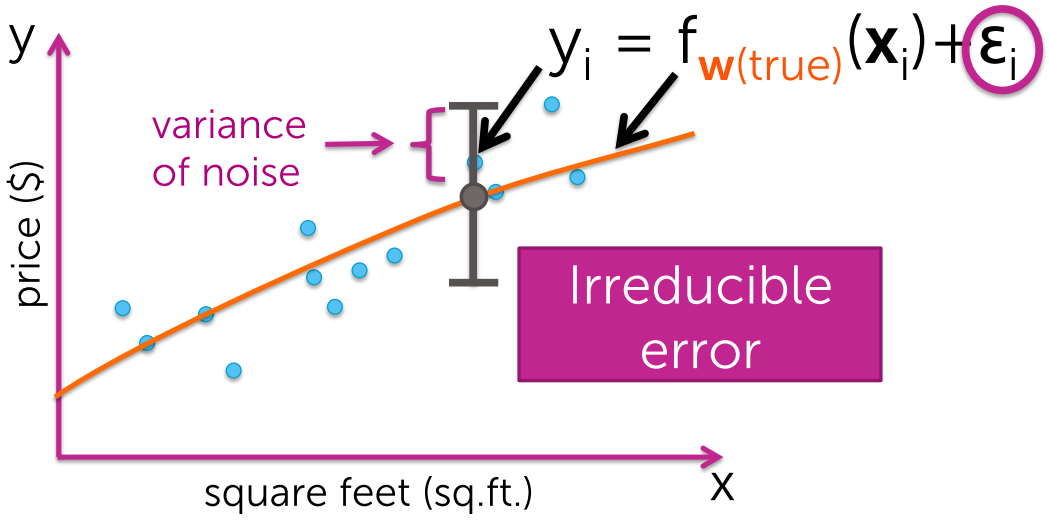
3 sources of error

In forming predictions, there are 3 sources of error:

1. Noise
2. Bias
3. Variance

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Data inherently noisy

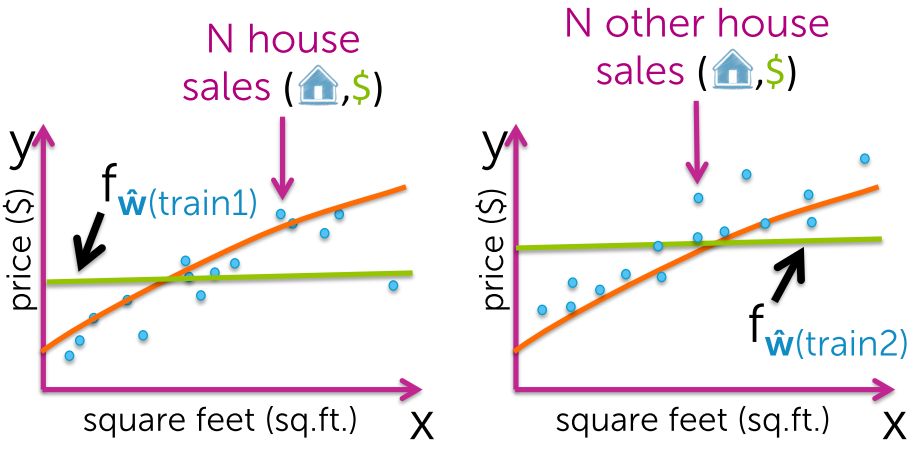


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Bias contribution

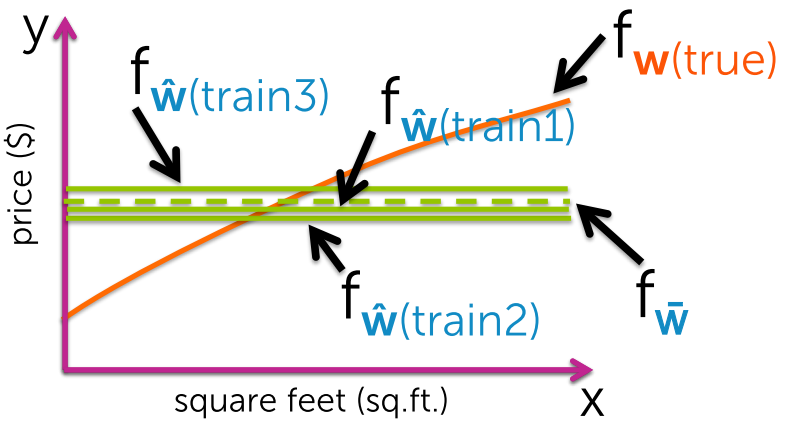
Assume we fit a constant function



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Bias contribution

Over all possible size N training sets, what do I expect my fit to be?

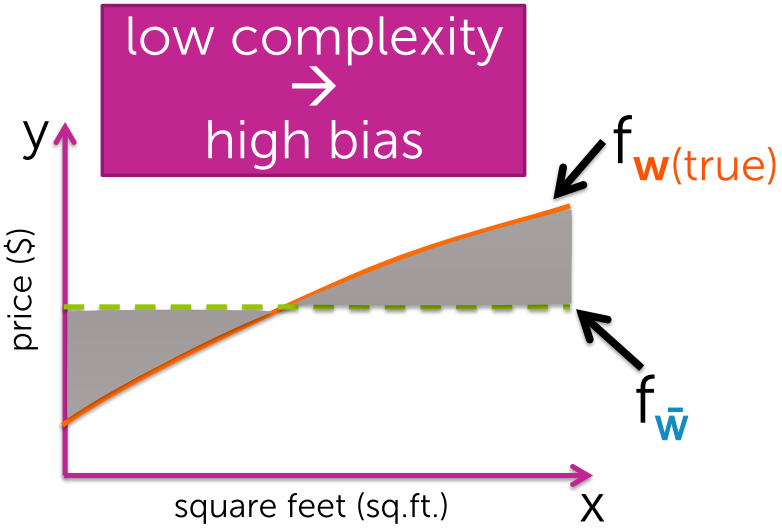


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Bias contribution

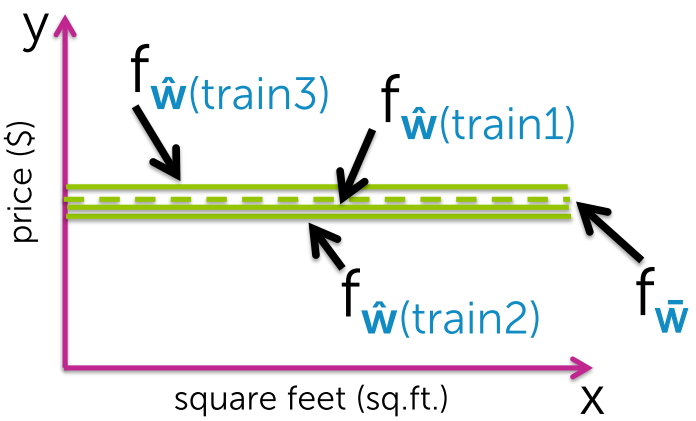
$\text{Bias}(\mathbf{x}) = f_{\mathbf{w}(\text{true})}(\mathbf{x}) - f_{\bar{\mathbf{w}}}(\mathbf{x})$ ← Is our approach flexible enough to capture $f_{\mathbf{w}(\text{true})}$? **If not, error in predictions.**



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Variance contribution

How much do specific fits vary from the expected fit?

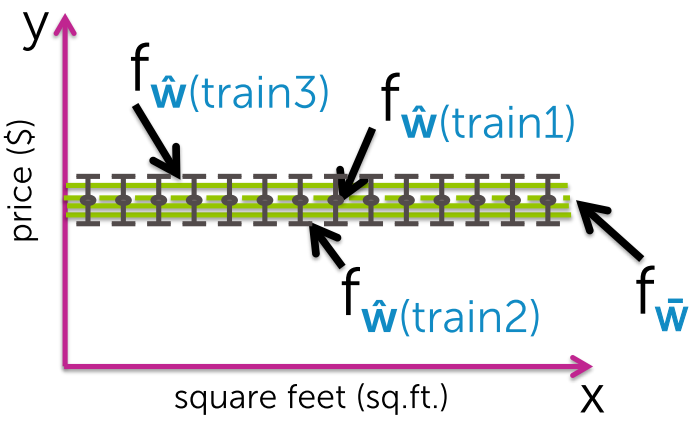


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Variance contribution

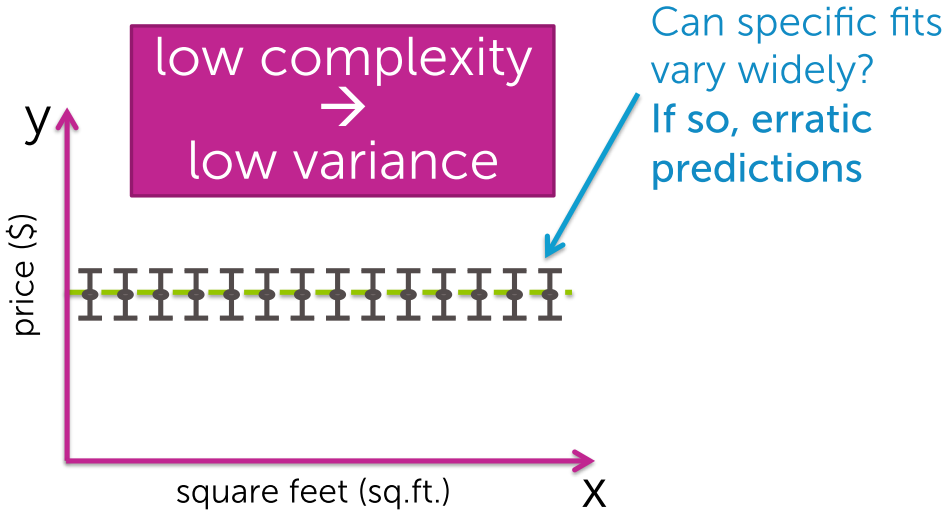
How much do specific fits vary from the expected fit?



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Variance contribution

How much do specific fits vary from the expected fit?

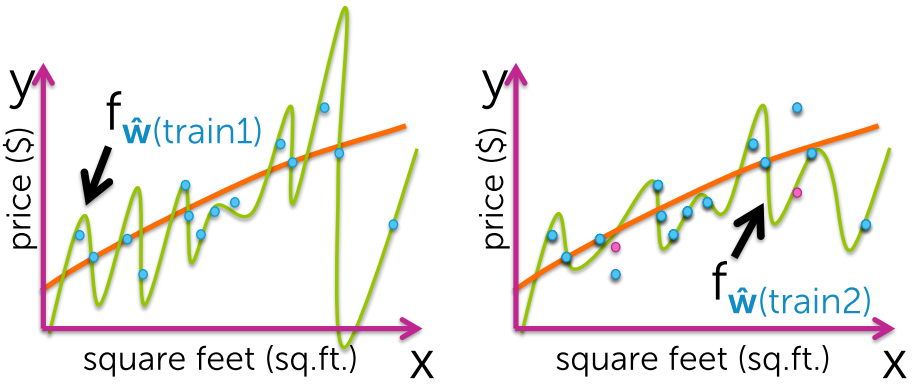


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Variance of high-complexity models

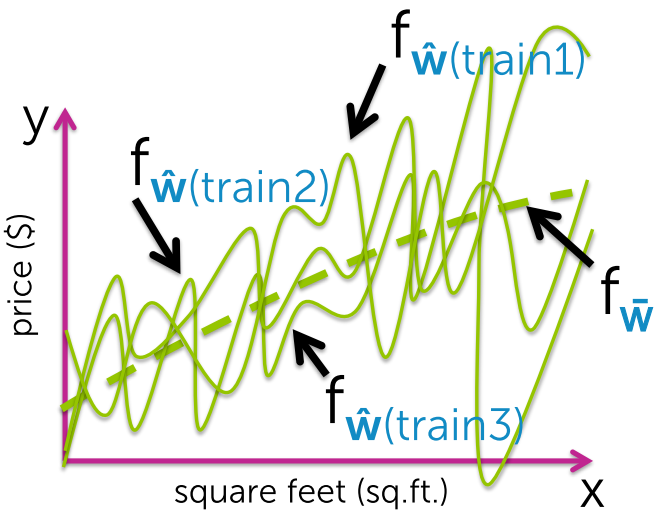
Assume we fit a high-order polynomial



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Variance of high-complexity models

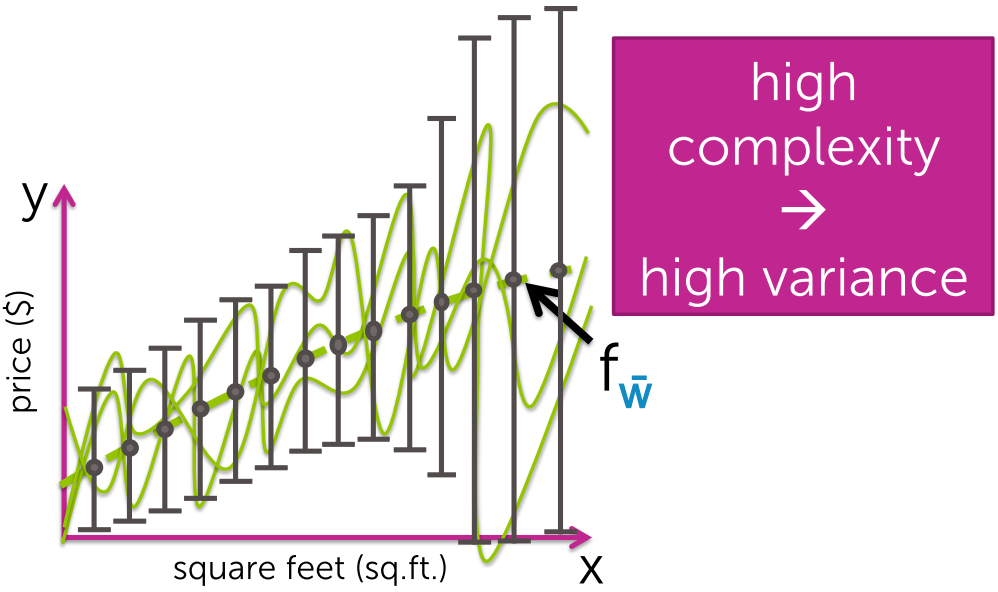
Assume we fit a high-order polynomial



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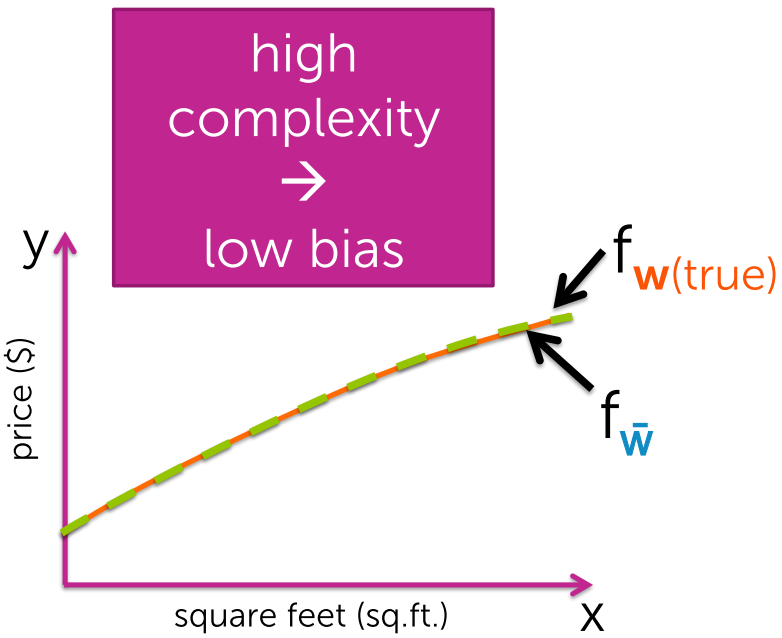
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Variance of high-complexity models



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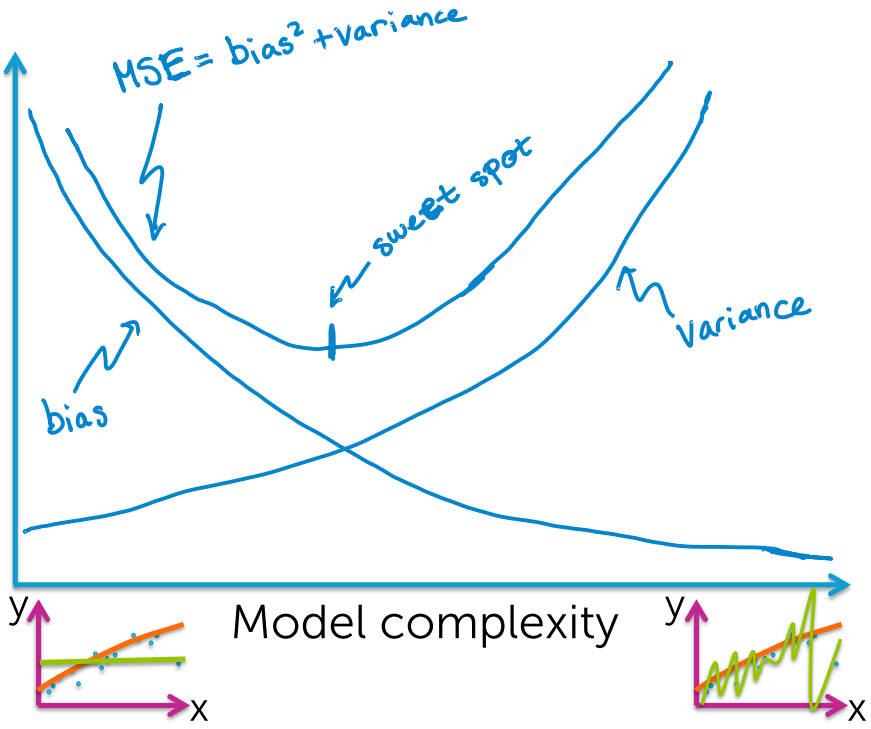
Bias of high-complexity models



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Bias-variance tradeoff

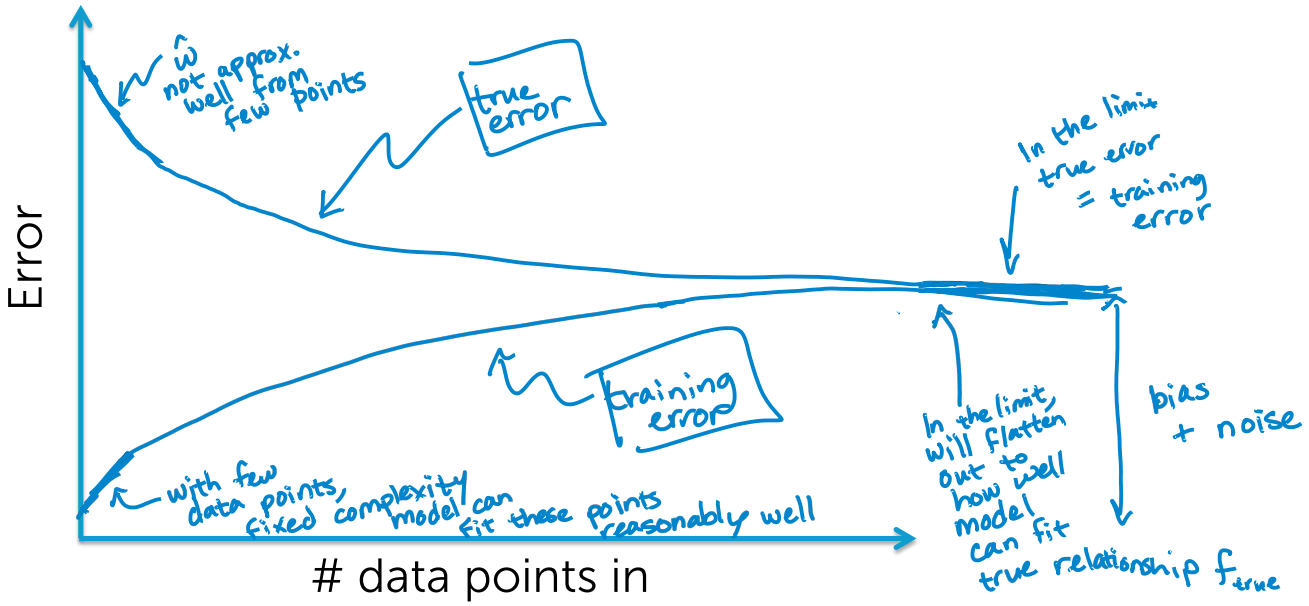


Just like with generalization error, we cannot compute bias and variance

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Error vs. amount of data

for a fixed model complexity



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More in depth on the
3 sources of errors...

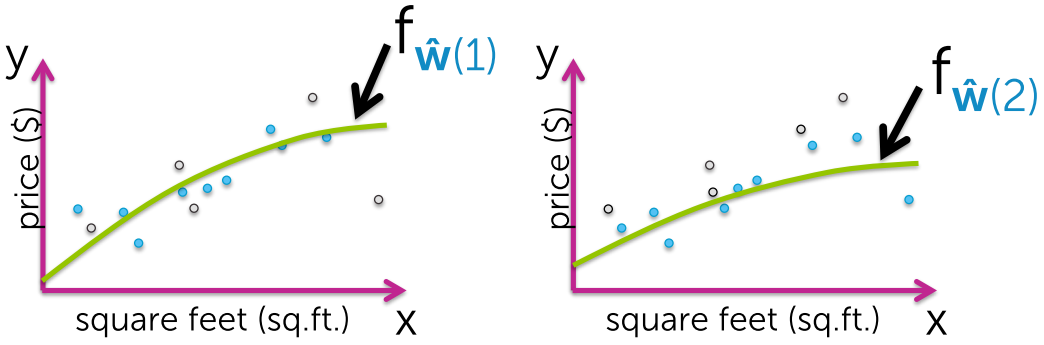


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Accounting for training set randomness

Training set was just a random sample of N houses sold

What if N other houses had been sold and recorded?



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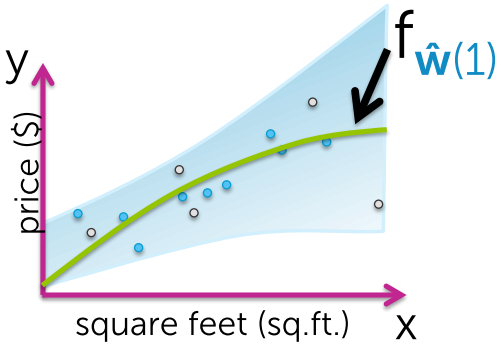
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Accounting for training set randomness

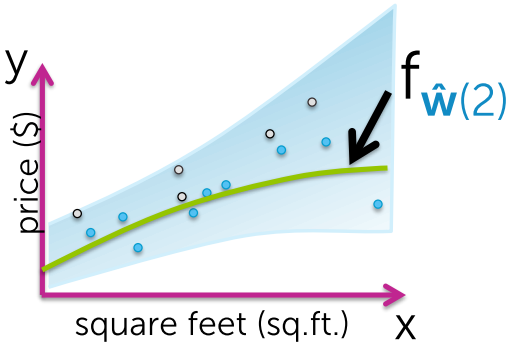
Training set was just a random sample of N houses sold

What if N other houses had been sold and recorded?

generalization error of $\hat{\mathbf{w}}(1)$



generalization error of $\hat{\mathbf{w}}(2)$

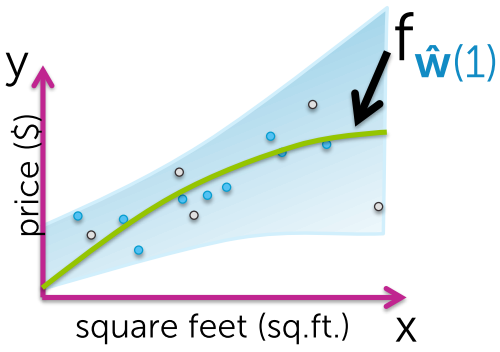


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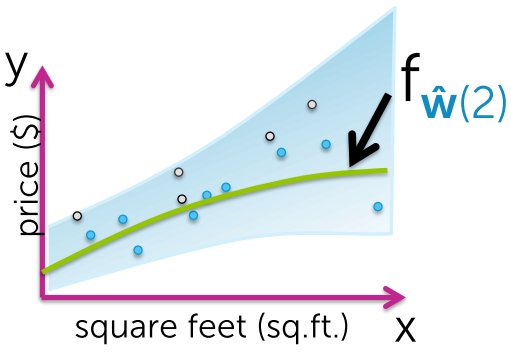
Accounting for training set randomness

Ideally, want performance averaged over all possible training sets of size N

generalization error of $\hat{\mathbf{w}}(1)$



generalization error of $\hat{\mathbf{w}}(2)$



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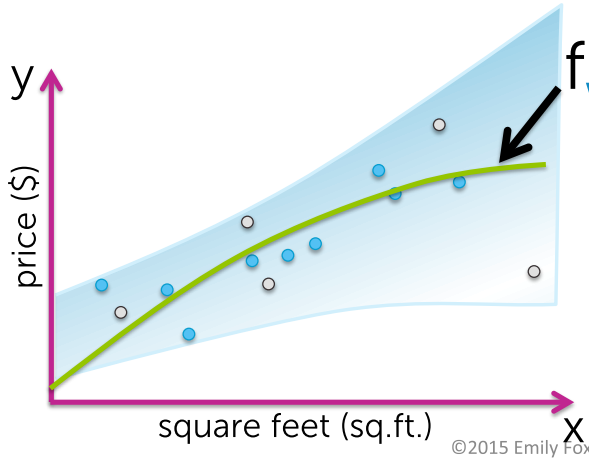
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Expected prediction error

$E_{\text{training set}}$[generalization error of $\hat{\mathbf{w}}$(training set)]

averaging over all training sets (weighted by how likely each is)

parameters fit on a specific training set

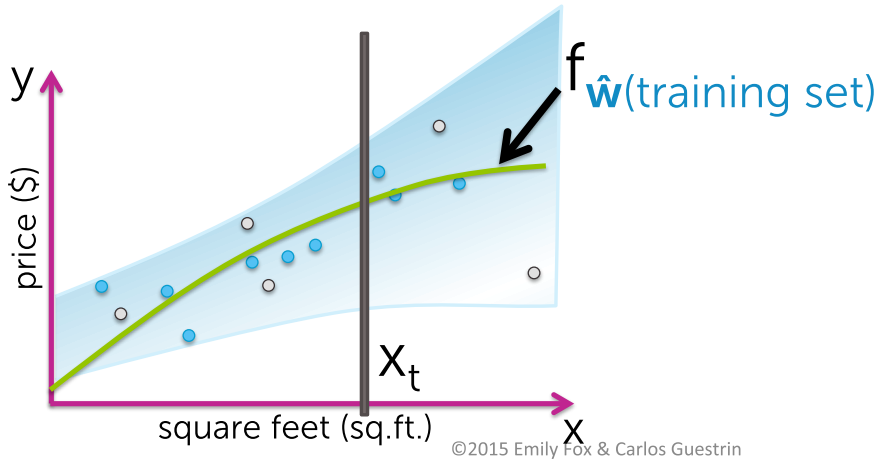


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Prediction error at target input

Start by considering:

1. Loss at target $\mathbf{x}_t$ (e.g. 2640 sq.ft.)
2. Squared error loss $L(y, f_{\hat{\mathbf{w}}}(\mathbf{x})) = (y - f_{\hat{\mathbf{w}}}(\mathbf{x}))^2$



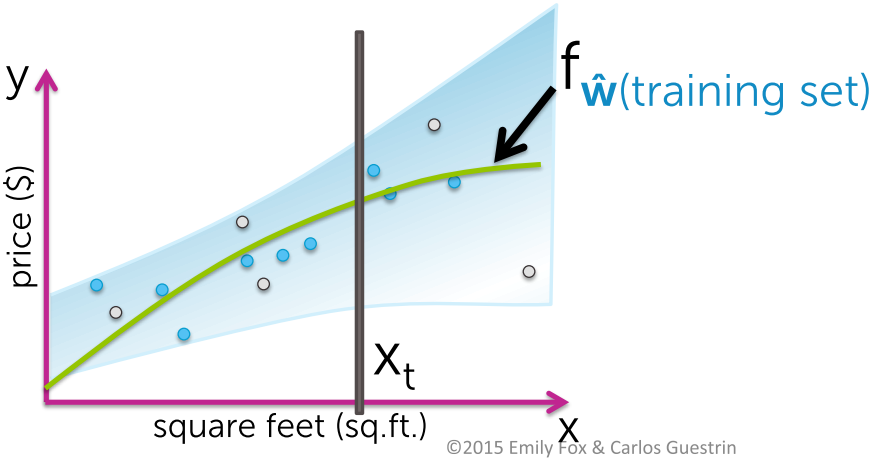
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Sum of 3 sources of error

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$



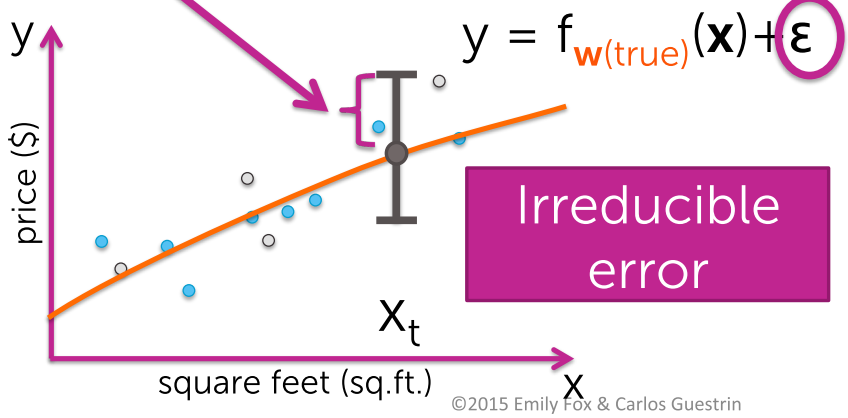
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Error variance of the model

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$

σ^2 = "variance"



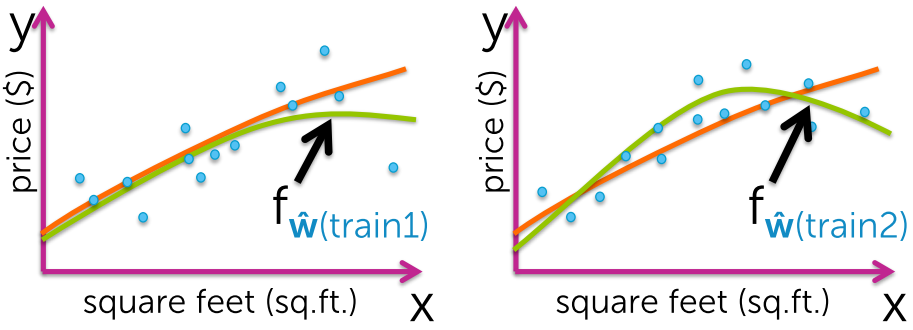
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Bias of function estimator

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$



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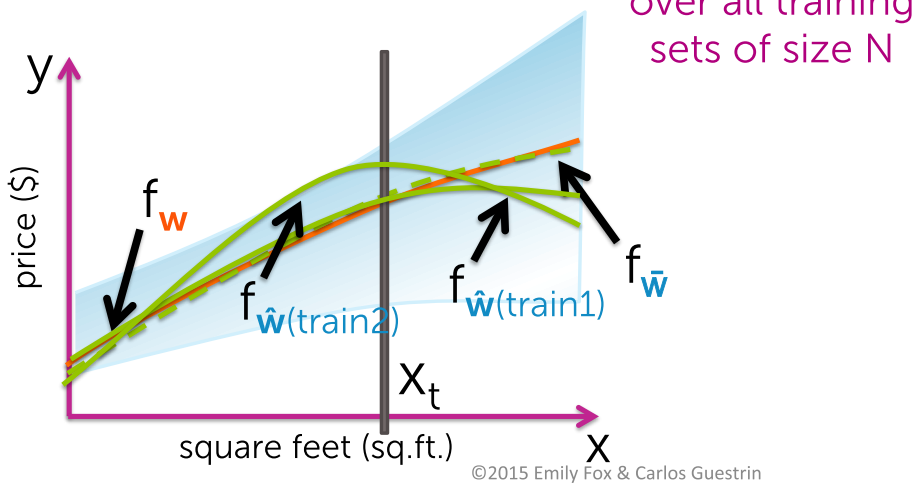
Bias of function estimator

Average estimated function = $f_{\bar{\mathbf{w}}}(\mathbf{x})$

True function = $f_{\mathbf{w}}(\mathbf{x})$

$E_{train}[f_{\hat{\mathbf{w}}(train)}(\mathbf{x})]$

over all training sets of size N



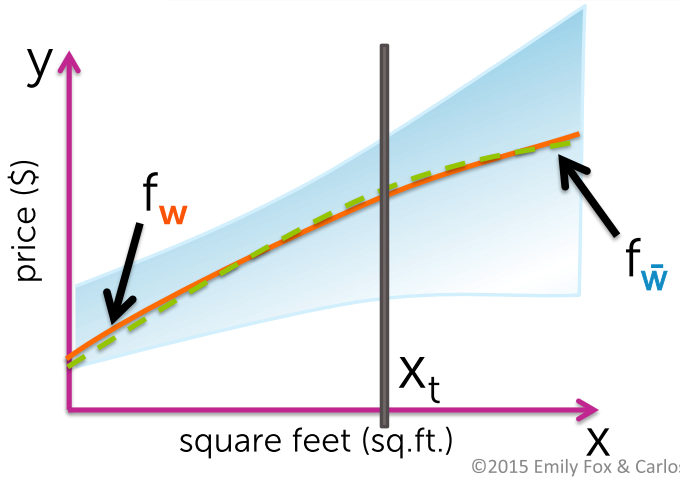
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Bias of function estimator

Average estimated function = $f_{\bar{w}}(\mathbf{x})$
True function = $f_{w}(\mathbf{x})$

$$\text{bias}(f_{\hat{w}}(\mathbf{x}_t)) = f_{w}(\mathbf{x}_t) - f_{\bar{w}}(\mathbf{x}_t)$$



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Bias of function estimator

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$

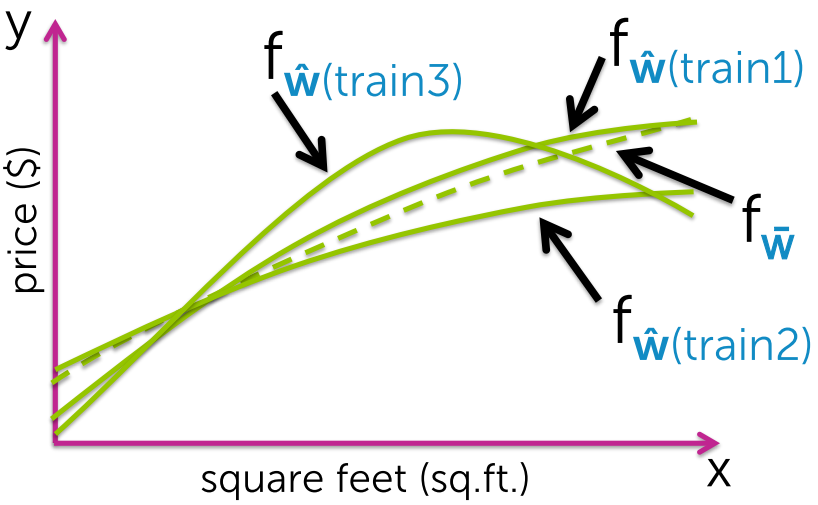
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Variance of function estimator

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$

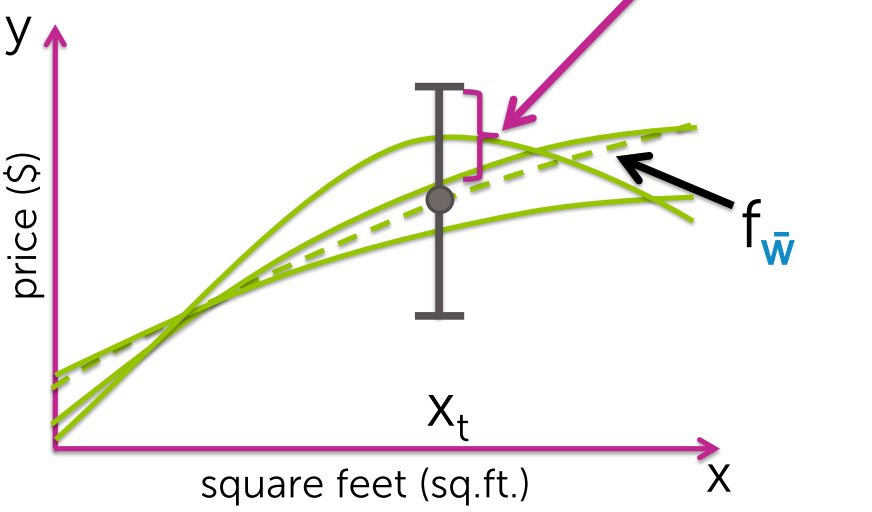


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Variance of function estimator

Average prediction error at $\mathbf{x}_t$

$$= \sigma^2 + [\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$



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Variance of function estimator

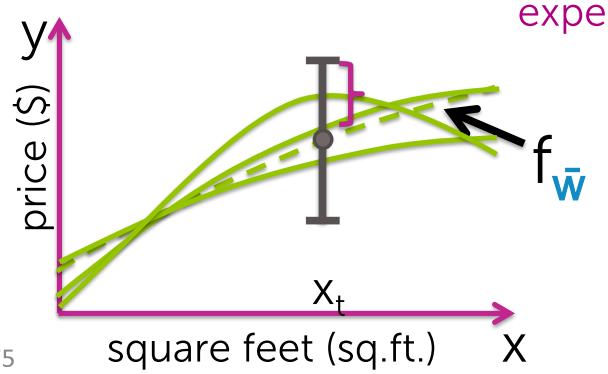
fit on a specific training dataset

what I expect to learn over all training sets

$$\text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t)) = E_{\text{train}}[(f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t) - f_{\bar{\mathbf{w}}}(\mathbf{x}_t))^2]$$

over all training sets of size N

deviation of specific fit from expected fit at $\mathbf{x}_t$



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Why 3 sources of error?
A formal derivation



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Deriving expected prediction error

Expected prediction error

$= E_{train}$ [generalization error of $\hat{\mathbf{w}}(train)$]

$= E_{train}\left[E_{\mathbf{x},y}\left[L(y, f_{\hat{\mathbf{w}}(train)}(\mathbf{x}))\right]\right]$

1. Look at specific $\mathbf{x}_t$
2. Consider $L(y, f_{\hat{\mathbf{w}}}(\mathbf{x})) = (y - f_{\hat{\mathbf{w}}}(\mathbf{x}))^2$

Expected prediction error at $\mathbf{x}_t$

$= E_{train, y_t}\left[(y_t - f_{\hat{\mathbf{w}}(train)}(\mathbf{x}_t))^2\right]$

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Deriving expected prediction error

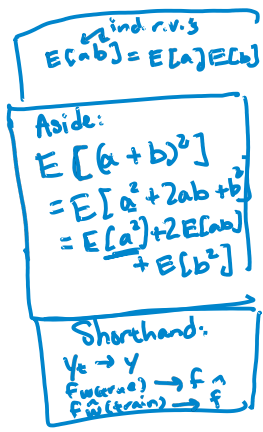
Expected prediction error at $\mathbf{x}_t$

$$= E_{\text{train}, y_t}\left[(y_t - f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t))^2\right]$$

$$= E_{\text{train}, y_t}\left[((y_t - f_{\mathbf{w}(\text{true})}(\mathbf{x}_t)) + (f_{\mathbf{w}(\text{true})}(\mathbf{x}_t) - f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t)))^2\right]$$

$$= E_{\text{train}, y}\left[(y-f)^2\right] + 2E_{\text{train}, y}\left[(y-f)(f-\hat{f})\right] + E_{\text{train}}\left[(f-\hat{f})^2\right]$$

$$= \sigma^2 + MSE(\hat{f})$$



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Equating MSE with bias and variance

$$\text{MSE}\left[f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t)\right] = E_{\text{train}}\left[\left(f_{\mathbf{w}(\text{true})}(\mathbf{x}_t) - f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t)\right)^2\right]$$

$$= E_{\text{train}}\left[\left(\left(f_{\mathbf{w}(\text{true})}(\mathbf{x}_t) - f_{\bar{\mathbf{w}}}(\mathbf{x}_t)\right) + \left(f_{\bar{\mathbf{w}}}(\mathbf{x}_t) - f_{\hat{\mathbf{w}}(\text{train})}(\mathbf{x}_t)\right)\right)^2\right]$$

$$= E_{\text{train}}\left[(f - \bar{f})^2\right] + 2E_{\text{train}}\left[(f - \bar{f})(\bar{f} - \hat{f})\right] + E_{\text{train}}\left[(\bar{f} - \hat{f})^2\right]$$

$(f - \bar{f})^2$ by definition $= \text{bias}^2(\hat{f})$

$2(f - \bar{f})\,E_{\text{train}}\left[\bar{f} - \hat{f}\right]$ — $(f - \bar{f})$ not a fcn of training data; $\bar{f} - E_{\text{train}}[\hat{f}]$ by definition $= 0$

$E\left[(\hat{f} - \bar{f})^2\right] = \text{var}(\hat{f})$ — $\hat{f}$: random function at x_t = random variable; $\bar{f}$: its mean

$$= \text{bias}^2(\hat{f}) + \text{var}(\hat{f})$$

$\bar{f}$ ← shorthand new notation on this slide

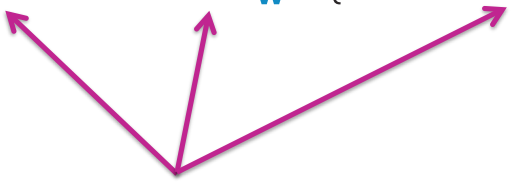
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Putting it all together

Expected prediction error at $\mathbf{x}_t$

$$= \sigma^2 + \text{MSE}\left[f_{\hat{\mathbf{w}}}(\mathbf{x}_t)\right]$$

$$= \sigma^2 + \left[\text{bias}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))\right]^2 + \text{var}(f_{\hat{\mathbf{w}}}(\mathbf{x}_t))$$



3 sources of error

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